CS23532-COMPUTER NETWORKS LAB MANUAL

EXPERIMENT-1

Basic Networking Commands in Linux and Windows operating systems

Aim:

To study and execute basic networking commands in Windows and Linux operating systems for viewing network configuration, testing connectivity, checking routing information, monitoring network status, and troubleshooting network-related issues.

Software Requirements

- Windows 10/11 or Linux (Ubuntu 20.04/22.04)
- Command Prompt (Windows)
- Terminal (Linux)
- Active Network Connection (LAN/Wi-Fi)

- **ipconfig / ifconfig / ip** – Displays network configuration.
- **ping** – Tests connectivity to a remote host.
- **tracert / traceroute** – Displays the path packets take to reach a destination.
- **nslookup** – Resolves domain names to IP addresses.
- **netstat** – Displays active network connections and listening ports.
- **arp** – Displays the ARP cache.
- **hostname** – Displays the system hostname.
- **route** – Displays or modifies the routing table

WINDOWS COMMANDS

Step 1: Open Command Prompt

1. Click **Start**.
2. Type **cmd**.
3. Click **Command Prompt**.

OR

Press

```
Windows + R
```

Type

```
cmd
```

Press **Enter**.

Command Prompt opens.

Command

```
cmd
ipconfig
```

Purpose

Displays

- IP Address
- Subnet Mask
- Default Gateway

Sample Output

```
IPv4 Address . . . . . : 192.168.1.20

Subnet Mask . . . . . : 255.255.255.0

Default Gateway . . . : 192.168.1.1
```

Step 3: Display Detailed Network Information

Command

```cmd
ipconfig /all
```

Shows

- MAC Address
- DHCP Server
- DNS Server
- Lease Information
- Adapter Details

Step 4: Test Network Connectivity

Command

```cmd
ping google.com
```

OR

```cmd
ping 8.8.8.8
```

Purpose

Checks whether the destination is reachable.

Sample Output

```
Reply from 8.8.8.8

Packets: Sent = 4

Received = 4

Lost = 0
```

Step 5: Trace Route to Destination

Command

```cmd
tracert google.com
```

Purpose

Displays every router (hop) between your computer and destination.

Sample Output

```
1    192.168.1.1

2    ISP Router

3    Google Network
```

Step 6: Display DNS Information

Command

```cmd
nslookup google.com
```

Purpose

Displays

- DNS Server
- IP Address of website

Example

```
Name: google.com

Address: 142.250.xxx.xxx
```

Step 7: Display Routing Table

Command

```cmd
route print
```

Purpose

Displays

- Network Routes
- Gateway
- Destination Networks

Step 8: Display Active Connections

Command

```cmd
netstat
```

Purpose

Displays

- Active TCP Connections
- Listening Ports

Step 9: Display Network Statistics

Command

```cmd
netstat -a
```

Shows

- All ports
- TCP connections
- UDP ports

Step 10: Display MAC Address

Command

```cmd
getmac
```

Purpose

Displays MAC address of the computer.

Step 11: Check Hostname

Command

```
cmd
hostname
```

Purpose

Displays computer name.

Example

```
DESKTOP-ABC123
```

Step 12: View ARP Table

Command

```
cmd
arp -a
```

Purpose

Displays IP-to-MAC address mappings.

Step 9: Display Network Statistics

Command

```
cmd
netstat -a
```

Shows

- All ports
- TCP connections
- UDP ports

Step 10: Display MAC Address

Command

```
cmd
getmac
```

Purpose

Displays MAC address of the computer.

LINUX NETWORKING COMMANDS

Step 1: Open Terminal

Press

```
Ctrl + Alt + T
```

OR

Search for **Terminal**.

Step 2: Display IP Address

Command

```bash
ip addr
```

OR

```bash
ifconfig
```

(if installed)

Purpose

Displays

Step 3: Display Routing Table

Command

```bash
ip route
```

Purpose

Shows

- Default Gateway
- Routing Table

Step 4: Display Hostname

Command

```bash
hostname
```

Purpose

Displays system hostname.

Step 5: Test Connectivity

Command

```bash
ping google.com
```

Stop using

```
Ctrl + C
```

Step 6: Trace Network Route

Command

```bash
traceroute google.com
```

If not installed

```bash
sudo apt install traceroute
```

Step 7: DNS Lookup

Command

```bash
nslookup google.com
```

If not installed

```bash
sudo apt install dnsutils
```

Step 8: Display Network Statistics

Command

```bash
netstat -a
```

OR

```bash
ss -a
```

Step 9: Display ARP Table

Command

```bash
arp -a
```

OR

```bash
ip neigh
```

Step 10: Display MAC Address

Command

```bash
ip link
```

Step 10: Display MAC Address

Command

```bash
ip link
```

OR

```bash
ifconfig
```

Look for

```
link/ether
```

Step 11: Display DNS Configuration

Command

```bash
cat /etc/resolv.conf
```

Shows

```
nameserver 8.8.8.8
```

Step 12: Check Interface Status

Command

```bash
ip link show
```

Displays

- UP/DOWN status
- Interface name

Step 13: Display Listening Ports

Command

```bash
ss -tuln
```

Purpose

Displays

- TCP ports
- UDP ports
- Listening services

Step 14: Check Internet Connectivity

Command

```bash
curl google.com
```

OR

```bash
wget google.com
```

EXPERIMENT -2,3 STUDY

EXPERIMENT-4 Implement packet sniffing using raw sockets in python

Aim

To implement a packet sniffer using **Python Raw Sockets** to capture and analyze network packets.

Requirements

Hardware

- Laptop/Desktop
- Minimum 4 GB RAM

Software

- Python 3.8 or above
- Windows 10/11 or Ubuntu Linux
- Visual Studio Code (VS Code) / IDLE / PyCharm
- Administrator (Windows) or Root (Linux) privileges

Procedure

Step 1: Install Python

1. Download Python from **https://www.python.org/downloads/**
2. Run the installer.
3. Select **Add Python to PATH**.
4. Click **Install Now**.
5. Wait until installation completes.





Step 4: Write the Python Program

Open VS Code or any Python editor.

Copy and save the following program as **packet_sniffer.py**.

import socket

Create raw socket

try:

s = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)

```
except PermissionError:
    print("Run this program as Administrator (Windows) or root (Linux).")
    exit()

# Bind to local IP address
HOST = "127.0.0.1"   # Replace with your local IP if needed
s.bind((HOST, 0))

# Include IP headers
s.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)

# Enable promiscuous mode (Windows only)
try:
    s.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
except AttributeError:
    pass

print("Packet Sniffer Started...\n")

try:
    while True:
        packet, addr = s.recvfrom(65565)
        print("Packet Received")
        print("Source IP:", addr[0])
        print("Packet Size:", len(packet), "bytes")
        print("-" * 50)
```

```
except KeyboardInterrupt:
    print("\nStopping Packet Sniffer...")
    try:
        s.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    except AttributeError:
        pass
    print("Packet Sniffer Stopped.")
```

Step 5: Find Your Local IP Address

Windows

Open Command Prompt:

```
cmd
ipconfig
```

Example:

```
IPv4 Address . . . . . : 192.168.1.10
```

Replace:

```
Python
HOST="127.0.0.1"
```

with

```
Python
HOST="192.168.1.10"
```

Linux

Run

```
Bash
ip addr
```

or

```bash
hostname -I
```

Use the displayed IP address.

Step 6: Run the Program

Windows

1. Open **Command Prompt as Administrator**.
2. Navigate to the project folder:

```cmd
cd Desktop\PacketSniffer
```

3. Execute:

```cmd
python packet_sniffer.py
```

Linux

Open Terminal and run:

```bash
sudo python3 packet_sniffer.py
```

Step 7: Generate Network Traffic

While the program is running, generate some network activity.

Open a browser

Visit:

```
https://www.google.com
```

OR Ping a Website

Windows:

```cmd
ping google.com
```

Linux:

```bash
ping google.com
```

Step 8: Observe the Output

Example:

```
Packet Sniffer Started...

Packet Received
Source IP: 192.168.1.5
Packet Size: 74 bytes
--------------------------------------------------

Packet Received
Source IP: 142.250.193.110
Packet Size: 98 bytes
--------------------------------------------------

Packet Received
Source IP: 192.168.1.5
Packet Size: 150 bytes
```

Step 9: Stop the Program

Press:

`Ctrl + C.`

EXPERIMENT-5 Develop a customized ping command to test the server connectivity

Aim

To develop a customized **Ping** program in Python that tests server connectivity by sending ICMP echo requests and displaying the response time and status.

Requirements

Hardware

- Laptop/Desktop
- Internet or Local Area Network (LAN)

Software

- Python 3.8 or above
- Visual Studio Code (VS Code) / IDLE / PyCharm
- Windows 10/11 or Ubuntu Linux.

Procedure

Step 1: Install Python

1. Download Python from **https://www.python.org/downloads/**
2. Run the installer.
3. Select **Add Python to PATH**.
4. Click **Install Now**.
5. Wait until installation completes.

Step 2: Verify Python Installation

Windows

Open **Command Prompt** and type:

```
cmd
python --version
```

Linux

Open **Terminal** and type:

```
Bash
python3 --version
```

Expected Output:

```
Python 3.x.x
```

Step 3: Create a Project Folder

1. Create a folder named:

```
CustomPing
```

2. Inside the folder, create a Python file named:

```
custom_ping.py
```

STEP 4:

```
import subprocess
import platform


host = input("Enter the server name or IP address: ")


if platform.system().lower() == "windows":
    command = ["ping", "-n", "4", host]
else:
    command = ["ping", "-c", "4", host]
```

```
print("\nTesting connectivity...\n")

result = subprocess.run(command, capture_output=True, text=True)

print(result.stdout)

if result.returncode == 0:
    print("Server is Reachable.")
else:
    print("Server is Not Reachable.")
```

Save the file.

Step 5: Open Command Prompt/Terminal

Windows

1. Press **Windows + R**.
2. Type:

```
cmd
```

3. Press **Enter**.

Linux

Press:

```
Ctrl + Alt + T
```

to open the Terminal.

Step 6: Navigate to the Project Folder

Windows

```cmd
cd Desktop\CustomPing
```

Linux

```Bash
cd ~/Desktop/CustomPing
```

Step 7: Execute the Program

Windows

```cmd
python custom_ping.py
```

Linux

```Bash
python3 custom_ping.py
```

Step 8: Enter the Server Address

Example:

```
Enter the server name or IP address:
google.com
```

or

```
8.8.8.8
```

Press **Enter**.

Step 9: Observe the Output

Example Output:

```
Enter the server name or IP address:
google.com

Testing connectivity...

Pinging google.com [142.250.183.46] with 32 bytes of
Reply from 142.250.183.46: bytes=32 time=18ms TTL=11
Reply from 142.250.183.46: bytes=32 time=17ms TTL=11
Reply from 142.250.183.46: bytes=32 time=19ms TTL=11
Reply from 142.250.183.46: bytes=32 time=18ms TTL=11

Ping statistics:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Server is Reachable.
```

Step 10: Test Another Server

Run the program again.

Example:

```
Enter the server name or IP address:
192.168.1.100
```

Observe whether the server is reachable.

Experiment 6: Building an Anonymous FTP Scanner Using Python ftplib Module

Aim

To develop an **Anonymous FTP Scanner** using Python's `ftplib` module to identify FTP servers that allow anonymous login.

Requirements

Hardware

- Laptop/Desktop
- Minimum 4 GB RAM
- Internet Connection

Software

- Windows 10/11 or Ubuntu Linux
- Python 3.8 or above
- Visual Studio Code (VS Code) / IDLE / PyCharm

Procedure

Step 1: Install Python

1. Download Python from **https://www.python.org/downloads/**.
2. Run the installer.
3. Select **Add Python to PATH**.
4. Click **Install Now**.
5. Wait for the installation to complete.

Step 2: Verify Python Installation

Windows

Open **Command Prompt** and execute:

```cmd
python --version
```

Linux

Open **Terminal** and execute:

```Bash
python3 --version
```

Expected Output

```
Python 3.x.x
```

Step 3: Create a Project Folder

1. Create a folder named:

```
FTPScanner
```

2. Inside the folder, create a Python file named:

```
anonymous_ftp_scanner.py
```

Step 4: Write the Python Program

Open **VS Code** or any Python editor.

Copy and save the following code in **anonymous_ftp_scanner.py**.

from ftplib import FTP

server = input("Enter FTP Server Address: ")

```
try:
    ftp = FTP(server)
    ftp.login('anonymous', 'anonymous@example.com')

    print("\nAnonymous Login Successful!")
    print("Connected to:", server)
    print("Server Message:")
    print(ftp.getwelcome())

    print("\nDirectory Listing:")
    ftp.retrlines('LIST')

    ftp.quit()

except Exception as e:
    print("\nAnonymous Login Failed!")
    print("Reason:", e)
```

Save the program.

Step 5: Open Command Prompt / Terminal

Windows

1. Press **Windows + R**.
2. Type:

```
cmd
```

3. Press **Enter**.

Linux

Press:

```
Ctrl + Alt + T
```

Step 6: Navigate to the Project Folder

Windows

```cmd
cd Desktop\FTPScanner
```

Linux

```bash
cd ~/Desktop/FTPScanner
```

Step 7: Run the Program

Windows

```cmd
python anonymous_ftp_scanner.py
```

Linux

```bash
python3 anonymous_ftp_scanner.py
```

Step 8: Enter the FTP Server Address

Example:

```
Enter FTP Server Address:
ftp.example.com
```

Press **Enter**.

Step 9: Observe the Output

Case 1: Anonymous Login Successful

```
Enter FTP Server Address:
ftp.example.com

Anonymous Login Successful!

Connected to: ftp.example.com

Server Message:
220 FTP Server Ready

Directory Listing:
drwxr-xr-x public
-rw-r--r-- readme.txt
```

Case 2: Anonymous Login Failed

```
Enter FTP Server Address:
ftp.example.com

Anonymous Login Failed!

Reason:
530 Login authentication failed.
```

Experiment 7: Develop a Simple Calculator Using XML-RPC.

Aim

To develop a **Simple Calculator** using the **XML-RPC (XML Remote Procedure Call)** protocol in Python to perform basic arithmetic operations between a client and a server.

Requirements

Hardware

- Laptop/Desktop
- Minimum 4 GB RAM

Software

- Windows 10/11 or Ubuntu Linux
- Python 3.8 or above
- Visual Studio Code (VS Code) / IDLE / PyCharm

Procedure

Step 1: Install Python

1. Download Python from **https://www.python.org/downloads/**
2. Install Python.
3. Select **Add Python to PATH** during installation.
4. Click **Install Now**.
5. Wait until the installation completes.



Step 3: Create a Project Folder

Create a folder named:

```
XMLRPC_Calculator
```

Inside the folder create two Python files:

```
server.py
```

```
client.py
```

Step 4: Write the Server Program

Open **server.py** and type the following code:

```
from xmlrpc.server import SimpleXMLRPCServer


# Create server
server = SimpleXMLRPCServer(("localhost", 8000))


print("XML-RPC Calculator Server Running on Port 8000...")


# Functions
def add(a, b):
    return a + b


def subtract(a, b):
    return a - b
```

```
def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return "Division by zero is not allowed."
    return a / b

# Register functions
server.register_function(add, "add")
server.register_function(subtract, "subtract")
server.register_function(multiply, "multiply")
server.register_function(divide, "divide")

# Start server
server.serve_forever()
```

Save the file.

Step 5: Write the Client Program

Open **client.py** and type the following code:

```
import xmlrpc.client

# Connect to server
server = xmlrpc.client.ServerProxy("http://localhost:8000")

a = int(input("Enter First Number: "))
```

```
b = int(input("Enter Second Number: "))

print("\nSelect Operation")
print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")

choice = input("Enter your choice (1-4): ")

if choice == '1':
    print("Result =", server.add(a, b))

elif choice == '2':
    print("Result =", server.subtract(a, b))

elif choice == '3':
    print("Result =", server.multiply(a, b))

elif choice == '4':
    print("Result =", server.divide(a, b))

else:
    print("Invalid Choice")
```

Save the file.

Step 6: Open Command Prompt/Terminal

Windows

Press

```
Windows + R
```

Type

```
cmd
```

Press **Enter**.

Linux

Press

```
Ctrl + Alt + T
```

Step 7: Navigate to the Project Folder

Windows

cmd

```
cd Desktop\XMLRPC_Calculator
```

Linux

Bash

```
cd ~/Desktop/XMLRPC_Calculator
```

Step 8: Start the XML-RPC Server

Run:

Windows

```cmd
python server.py
```

Linux

```bash
python3 server.py
```

Expected Output

```
XML-RPC Calculator Server Running on Port 8000...
```

Keep this terminal open.

Step 9: Open Another Command Prompt/Terminal

Open a **new** Command Prompt (Windows) or Terminal (Linux).

Navigate to the same project folder.

Step 10: Run the Client Program

Windows

```cmd
python client.py
```

Linux

```bash
python3 client.py
```

Step 11: Enter the Input

Example:

```
Enter First Number:
20

Enter Second Number:
10
```

Choose the operation.

```
1. Addition
2. Subtraction
3. Multiplication
4. Division

Enter your choice:
1
```

Step 12: Observe the Output

Addition

```
Enter First Number: 20
Enter Second Number: 10

Result = 30
```

Subtraction

```
Result = 10
```

Multiplication

```
Result = 200
```

Division

```
Result = 2.0
```

Experiment: 9 Design a Simple Topology and Configure One Router, Two Switches, and PCs Using Cisco Packet Tracer

Aim

To design and configure a simple network topology consisting of **one router, two switches, and four PCs** using **Cisco Packet Tracer**, and verify communication between all devices.

Software

- Cisco Packet Tracer (Version 8.x or later)

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router R1	G0/0	192.168.1.1	255.255.255.0	-
Router R1	G0/1	192.168.2.1	255.255.255.0	-
PC0	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC2	NIC	192.168.2.2	255.255.255.0	192.168.2.1
PC3	NIC	192.168.2.3	255.255.255.0	192.168.2.1

Procedure

Step 1: Open Cisco Packet Tracer

1. Click **Start**.
2. Open **Cisco Packet Tracer**.
3. Wait for the workspace to load.

Step 2: Add Devices

From the **Network Devices** section:

- Drag **1 Router** (Cisco 1941 or 2911).
- Drag **2 Switches** (Cisco 2960).
- Drag **4 PCs**.

Arrange the devices as shown in the topology.

Step 3: Connect the Devices

Use the **Copper Straight-Through Cable**.

Make the following connections:

From	To
Router G0/0	Switch1 Fa0/1
Router G0/1	Switch2 Fa0/1
PC0 FastEthernet0	Switch1 Fa0/2
PC1 FastEthernet0	Switch1 Fa0/3
PC2 FastEthernet0	Switch2 Fa0/2
PC3 FastEthernet0	Switch2 Fa0/3

Wait until the link indicators turn **green**.

Router Configuration

Step 4: Open Router CLI

1. Click the **Router**.
2. Select the **CLI** tab.
3. Press **Enter** when prompted.

Step 5: Configure Interface G0/0

Type the following commands:

Router> enable

Router# configure terminal

Router(config)# interface gigabitEthernet0/0

Router(config-if)# ip address 192.168.1.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit

Step 6: Configure Interface G0/1

Router(config)# interface gigabitEthernet0/1

Router(config-if)# ip address 192.168.2.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit

Step 7: Save Router Configuration

Router(config)# end

Router# copy running-config startup-config

Press **Enter** when prompted.

PC Configuration

Step 8: Configure PC0

1. Click **PC0**.
2. Select **Desktop**.
3. Click **IP Configuration**.
4. Enter:

Field	Value
IP Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

Close the window.

Step 9: Configure PC1

Field	Value
IP Address	192.168.1.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

Step 10: Configure PC2

Field	Value
IP Address	192.168.2.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1

Step 11: Configure PC3

Field	Value
IP Address	192.168.2.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1

Step 12: Test Connectivity

Test 1: PC0 to PC1

1. Click **PC0**.
2. Go to **Desktop → Command Prompt**.
3. Type:

   ping 192.168.1.3

Expected Output:

Reply from 192.168.1.3:

Packets: Sent = 4

Received = 4

Lost = 0

Test 2: PC0 to PC2

```
ping 192.168.2.2
```

Expected Output:

```
Reply from 192.168.2.2
```

Test 3: PC1 to PC3

```
ping 192.168.2.3
```

Expected Output:

```
Reply from 192.168.2.3
```

Test 4: PC3 to Router

```
ping 192.168.2.1
```

Expected Output:

```
Reply from 192.168.2.1
```

Expected Output

```
PC0> ping 192.168.2.2

Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Packets: Sent = 4
Received = 4
Lost = 0 (0% loss)
```

Experiment: 10 Customize Switch with Network Modules Using Cisco Packet Tracer

Aim

To customize a Cisco switch by adding or removing network modules using **Cisco Packet Tracer** and verify the changes in the switch interfaces.

Software

- **Cisco Packet Tracer Version 8.x or later**

Procedure

Step 1: Open Cisco Packet Tracer

1. Click **Start**.
2. Open **Cisco Packet Tracer**.
3. Wait until the workspace loads.

Step 2: Add a Modular Switch

1. Click **Network Devices**.
2. Select **Switches**.
3. Drag a **Cisco 3560** or another modular switch (if supported in your Packet Tracer version) onto the workspace.

Note: Some switch models (such as the Cisco 2960) are fixed-configuration switches and do **not** support adding modules. Use a modular switch model if available.

Step 3: Open the Physical View

1. Click the switch.
2. Select the **Physical** tab.
3. Observe the chassis and installed modules.

Step 4: Turn Off the Switch

1. Locate the **Power Switch** in the Physical tab.
2. Click it to turn the switch **OFF**.
3. The power indicator changes to **OFF**.

Important: Modules can only be inserted or removed while the switch is powered off.

Step 5: Remove an Existing Module (Optional)

1. Select the module installed in the expansion slot.

2. Drag it out of the slot if you want to replace it.

Step 6: Add a Network Module

1. From the list of available modules in the Physical tab, choose a compatible module (for example, an Ethernet expansion module if available).
2. Drag the module into an empty expansion slot.
3. Ensure the module fits correctly.

Step 7: Power On the Switch

1. Click the **Power Switch** again.
2. Wait until the switch boots completely.
3. Verify that the new module is detected.

Step 8: Verify the New Interfaces

1. Click the **Config** tab.
2. Observe the list of interfaces.
3. Confirm that the newly added ports are available.

Step 9: Connect End Devices

1. Drag **2 PCs** to the workspace.
2. Use **Copper Straight-Through** cables.
3. Connect:
    - **PC0 → Switch Port**
    - **PC1 → Switch Port**

Step 10: Configure IP Addresses

PC0

1. Click **PC0**.
2. Select **Desktop → IP Configuration**.
3. Enter:

Parameter	Value
IP Address	192.168.10.2
Subnet Mask	255.255.255.0

PC1

1. Click **PC1**.
2. Select **Desktop → IP Configuration**.
3. Enter:

Parameter	Value
IP Address	192.168.10.3
Subnet Mask	255.255.255.0

Step 11: Test Connectivity

1. Click **PC0**.
2. Open **Desktop → Command Prompt**.
3. Execute:

```
ping 192.168.10.3
```

Step 12: Observe the Output

Expected Output:

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Packets: Sent = 4

Received = 4

Lost = 0 (0% loss)

Step 13: Save the Packet Tracer File

1. Click **File → Save As**.
2. Enter a filename such as:

```
Customize_Switch_Modules.pkt
```

3. Click **Save**.

Experiment:11 Examine Network Address Translation (NAT) Using Cisco Packet Tracer

Aim

To configure and examine **Network Address Translation (NAT)** using **Cisco Packet Tracer** and verify communication between a private network and a public network.

Software

- Cisco Packet Tracer (Version 8.x or later)

Procedure

Step 1: Open Cisco Packet Tracer

1. Click **Start**.
2. Open **Cisco Packet Tracer**.
3. Wait until the workspace loads.

Step 2: Add Network Devices

Drag the following devices into the workspace:

- 1 Router (Cisco 1941 or 2911)
- 1 Switch (2960)
- 1 PC
- 1 Server

Arrange them as shown in the topology.

Step 3: Connect the Devices

Use **Copper Straight-Through** cables.

From	To
PC0 FastEthernet0	Switch Fa0/1
Switch Fa0/2	Router G0/0
Router G0/1	Server FastEthernet0

Wait until all links turn **green**.

Configure the Router

Step 4: Open Router CLI

1. Click **Router**.
2. Select the **CLI** tab.
3. Press **Enter**.

Step 5: Configure the Inside Interface.

Router> enable

Router# configure terminal

Router(config)# interface gigabitEthernet0/0

Router(config-if)# ip address 192.168.1.1 255.255.255.0

Router(config-if)# ip nat inside

Router(config-if)# no shutdown

Router(config-if)# exit

Step 6: Configure the Outside Interface

Router(config)# interface gigabitEthernet0/1

Router(config-if)# ip address 200.1.1.1 255.255.255.0

Router(config-if)# ip nat outside

Router(config-if)# no shutdown

Router(config-if)# exit

Step 7: Configure Static NAT

Map the private IP address **192.168.1.2** to the public IP address **200.1.1.100**.

Router(config)# ip nat inside source static 192.168.1.2 200.1.1.100

Step 8: Save the Configuration

```
Router(config)# end
Router# copy running-config startup-config
```

Press **Enter** to confirm.

Configure the PC

Step 9: Assign IP Address

1. Click **PC0**.
2. Select **Desktop → IP Configuration**.
3. Enter:

Parameter	Value
IP Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

Close the window.

Configure the Server

Step 10: Assign IP Address

1. Click **Server**.
2. Select **Desktop → IP Configuration**.
3. Enter:

Parameter	Value
IP Address	200.1.1.2
Subnet Mask	255.255.255.0
Default Gateway	200.1.1.1

Verify NAT

Step 11: Test Connectivity

1. Click **PC0**.
2. Select **Desktop → Command Prompt**.
3. Execute:

```
ping 200.1.1.2
```

Step 12: Verify NAT Translation

On the router CLI, enter:

```
Router# show ip nat translations
```

Expected Output

```
Pro Inside global      Inside local       Outside local     Outside global
200.1.1.100            192.168.1.2        ---               ---
```

Step 13: View NAT Statistics

Execute:

```
Router# show ip nat statistics
```

This displays:

- Number of active translations
- Inside interfaces
- Outside interfaces
- Translation statistics .

EXPERIMENT-12

Nmap to Discover Live Hosts Using Nmap (ARP Scan, ICMP Scan, TCP Ping Scan, and UDP Ping Scan) in Try Hack Me Platform.

Aim

To discover live hosts in a network using **Nmap** by performing:

- ARP Scan
- ICMP Ping Scan
- TCP Ping Scan
- UDP Ping Scan
  on the **TryHackMe** platform.

Requirements

- TryHackMe Account
- Internet Connection
- AttackBox (provided by TryHackMe) or Kali Linux VM
- Nmap installed
- Target machine deployed in TryHackMe

Procedure

Step 1: Login to TryHackMe

1. Open your browser.
2. Visit

https://tryhackme.com

3. Login using your account.

Step 2: Open the Required Room

Examples:

- Nmap Room
- Network Services Room
- Nmap Live Host Discovery Room

Step 3: Deploy the Machine

Click

start Machine

Wait until the machine starts.

Example:

`Target IP`

`10.10.210.15`

Step 4: Start the AttackBox

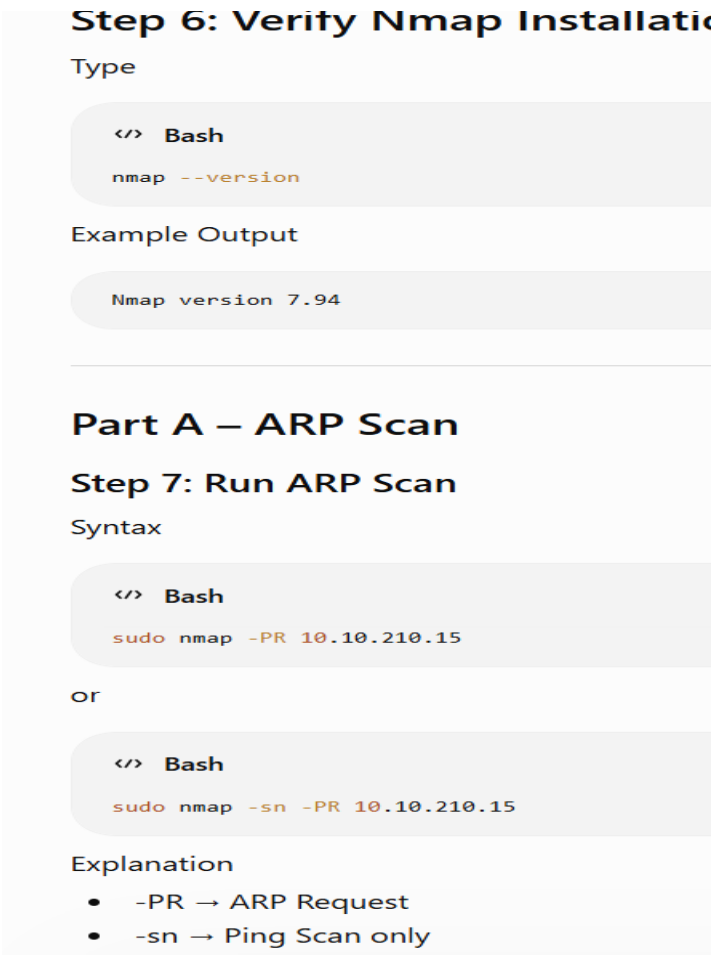
Click

Start AttackBox

Wait for the Kali Linux desktop to load.

Step 5: Open Terminal

Open Terminal from the AttackBox.



Sample Output

```
Starting Nmap

Nmap scan report for 10.10.210.15

Host is up.

MAC Address:
02:42:AC:11:00:02

Nmap done.
```

Observation

The target is alive because it responded to the ARP request.

Part B – ICMP Ping Scan

Step 8: Run ICMP Scan

Command

```bash
sudo nmap -PE 10.10.210.15
```

or

```bash
sudo nmap -sn -PE 10.10.210.15
```

Explanation

- -PE → ICMP Echo Request

Sample Output

```
Host is up (0.0010s latency)
```

Observation

The host replied to the ICMP Echo Request.

Part C – TCP SYN Ping Scan

Step 9: Perform TCP Ping Scan

Command

```bash
sudo nmap -PS22,80,443 10.10.210.15
```

Explanation

- -PS → TCP SYN Ping
- Sends SYN packets to ports 22, 80, and 443.

Sample Output

```
Host is up.

22/tcp filtered
80/tcp closed
443/tcp open
```

Part D – TCP ACK Ping Scan

Step 10: Run TCP ACK Ping

Command

```bash
sudo nmap -PA80 10.10.210.15
```

Explanation

- -PA → TCP ACK Ping

Sample Output

```
Host is up.
```

Part E – UDP Ping Scan

Step 11: Perform UDP Ping

Command

```bash
sudo nmap -PU53,161 10.10.210.15
```

Explanation

- -PU → UDP Ping
- Sends UDP packets to ports 53 and 161.

Sample Output

```
Host is up.
```

Step 12: Discover All Hosts in a Network

Example

```bash
sudo nmap -sn 10.10.210.0/24
```

Output

```
Nmap scan report for 10.10.210.5

Host is up.

Nmap scan report for 10.10.210.10

Host is up.

Nmap scan report for 10.10.210.15

Host is up.
```

Step 13: Compare Different Ping Techniques

Run the following commands one after another:

Bash

```
sudo nmap -sn -PR 10.10.210.15
```

Bash

```
sudo nmap -sn -PE 10.10.210.15
```

Bash

```
sudo nmap -sn -PS80 10.10.210.15
```

Bash

```
sudo nmap -sn -PA80 10.10.210.15
```

Bash

```
sudo nmap -sn -PU53 10.10.210.15
```

Observe which scan successfully detects the host.

Step 14: Save the Scan Results

Save in normal format:

Bash

```
sudo nmap -sn -PR 10.10.210.15 -oN arp_scan.txt
```

Save in XML format:

Bash

```
sudo nmap -sn -PE 10.10.210.15 -oX icmp_scan.xml
```

Save in all formats:

Bash

```
sudo nmap -sn -PS80 10.10.210.15 -oA tcp_ping
```

This creates:

```
tcp_ping.nmap
tcp_ping.xml
tcp_ping.gnmap
```

EXPERIMENT-13 Demonstrate network forensics using Pcap Xraytool

Aim

To perform **network forensics** by analyzing a packet capture (PCAP) file using the **PCAP X-Ray** tool to identify network protocols, suspicious activities, communication patterns, transferred files, and security indicators.

Software

- Windows/Linux Operating System
- Web Browser (Google Chrome, Firefox, or Edge)
- PCAP X-Ray (Online Tool)
- Sample PCAP file (from Wireshark, TryHackMe, or CyberDefenders)

Step 1: Obtain a PCAP File

Use one of the following methods:

Option 1: Capture Your Own Traffic

1. Open **Wireshark**.
2. Select the active network interface.
3. Click **Start Capture**.
4. Browse a few websites or perform simple network activities.
5. Stop the capture.
6. Save the file as:

```
sample.pcap
```

Option 2: Download a Sample PCAP

Download a PCAP file from:

- TryHackMe rooms
- CyberDefenders
- Malware Traffic Analysis (training datasets)

Step 2: Open the PCAP X-Ray Tool

1. Open a web browser.
2. Navigate to the **PCAP X-Ray** website.

3. The upload interface will appear.

Step 3: Upload the PCAP File

1. Click **Choose File**.
2. Select the saved **sample.pcap** file.
3. Click **Upload**.
4. Wait for the analysis to complete.

Step 4: View the Dashboard After processing, the dashboard displays:

- Total packets
- Total protocols
- Source IP addresses
- Destination IP addresses
- Communication statistics

Record these values in your observation table.

Step 5: Analyze Protocol Distribution

Open the **Protocols** section.

Observe:

- TCP
- UDP
- HTTP
- HTTPS
- DNS
- ARP
- ICMP
- SMTP (if present)
- FTP (if present)

Sample Observation

Protocol	Number of Packets
TCP	720
UDP	180
DNS	60

Protocol	Number of Packets
HTTP	140
HTTPS	320

Step 6: Examine Source and Destination Hosts

Open the **Hosts** or **Endpoints** section.

Record:

- Source IP addresses
- Destination IP addresses
- Number of packets exchanged
- Data transferred

Example:

Source IP	Destination IP	Packets
192.168.1.5	8.8.8.8	120
192.168.1.5	142.250.x.x	240

Step 7: Analyze DNS Queries

Navigate to the **DNS** section.

Observe:

- Domain names
- DNS server IP
- Query type (A, AAAA, MX, etc.)
- Response IP addresses

Example:

```
google.com

openai.com

github.com
```

Step 8: Inspect HTTP Requests

Open the **HTTP** section.

Observe:

- Request method (GET, POST)
- Requested URL
- User-Agent
- Response codes (200, 404, 301, etc.)

Example:

```
GET /index.html

Status: 200 OK

Host: example.com
```

Step 9: Review TLS/HTTPS Sessions

Open the **TLS/SSL** section.

Observe:

- Server Name Indication (SNI)
- TLS version
- Certificates (if available)
- Cipher suite

Example:

```
TLS Version:

TLS 1.3

Server:

github.com
```

Step 10: Inspect Conversations

Open the **Conversations** section.

Review:

- Client IP
- Server IP
- Port numbers
- Number of packets
- Total bytes exchanged

Example:

Client	Server	Protocol
192.168.1.5	8.8.8.8	UDP
192.168.1.5	142.250.x.x	TCP

Step 11: Check for Credentials

If supported, open the **Credentials** section.

Look for:

- FTP usernames/passwords
- HTTP Basic Authentication
- Telnet credentials
- SMTP login information

Note: Credentials will only appear if they were transmitted in plaintext.

Step 12: Extract Files

Open the **Files** section.

Identify any transferred:

- Images
- PDF documents
- ZIP files
- Executable files

Download them for further analysis if required.

Step 13: Identify Security Indicators

Review indicators such as:

- Connections to suspicious IP addresses
- Malware-related domains
- Unusual port usage
- Excessive DNS queries
- Large outbound data transfers

Record any suspicious observations.

Step 14: View the Timeline

Open the **Timeline** section.

Observe:

- First packet timestamp
- Last packet timestamp
- Sequence of events
- Duration of communication

Example:

```
10:15:20

DNS Query

↓

HTTP Request

↓

HTTPS Session

↓

File Download
```

Step 15: Generate the Report

1. Click **Export** or **Download Report** (if available).
2. Save the report in HTML or PDF format.
3. Store it for documentation

EXPERIMENT-14

To capture, save, and analyze network traffic on TCP/UDP/ IP/HTTP/ARP/DHCP/ICMP/DNS using Wireshark Tool.

Aim

To capture, save, and analyze network traffic of TCP, UDP, IP, HTTP, ARP, DHCP, ICMP, and DNS protocols using the Wireshark network protocol analyzer.

Software Requirements

- Operating System: Windows 10/11 or Ubuntu Linux
- Wireshark (Latest Version)
- Internet Connection
- Administrator Privileges
- Web Browser (Chrome/Firefox/Edge)
- Command Prompt/Terminal

Procedure

Step 1: Install Wireshark

1. Download Wireshark from the official website.
2. Install Wireshark.
3. During installation:
    - Install **Npcap** (Windows).
    - Select **Install WinPcap API Compatible Mode**.
4. Finish installation.
5. Restart the computer if required.

Step 2: Open Wireshark

1. Click **Start Menu**.
2. Open **Wireshark**.
3. The home screen displays all available network interfaces.

Example:

```
Wi-Fi
Ethernet
Bluetooth
Loopback
```

Step 3: Select the Network Interface

1. If using Wi-Fi, choose **Wi-Fi**.
2. If using LAN, choose **Ethernet**.
3. Double-click the active interface.

Packet capture begins immediately.

You will observe moving colored packets.

Step 4: Start Capturing Packets

Click the **Blue Shark Fin** icon.

Wireshark starts capturing live packets.

Observe packets like:

```
No.
Time
Source
Destination
Protocol
Length
Info
```

Step 5: Generate Network Traffic

To capture different protocols, perform the following actions.

Part A – Capture ICMP Packets

Generate Traffic

Open Command Prompt.

Type.

```
ping google.com
```

or

```
ping 8.8.8.8
```

Press Enter.

Apply Display Filter

```
icmp
```

Observe:

- Echo Request
- Echo Reply

Fields:

- Source IP
- Destination IP
- TTL
- Sequence Number

Part B – Capture DNS Packe

Open browser.

Visit

```
www.google.com
```

or

```
www.wikipedia.org
```

Apply filter

```
dns
```

Observe:

- Standard Query
- Standard Response
- Domain Name
- Response Address

Part C – Capture HTTP Packets

Open browser.

Visit

```
http://neverssl.com
```

(This site uses HTTP.)

Apply filter

```
http
```

Observe:

- HTTP GET
- HTTP Response
- Status Code
- Host Name
- User-Agent

Part D – Capture TCP Packets

Open any website.

Filter

```
tcp
```

Observe:

TCP Three-Way Handshake

```
SYN

SYN + ACK

ACK
```

Observe

- Source Port
- Destination Port
- Sequence Number
- Acknowledgement Number
- Window Size

Part E – Capture UDP Packets

UDP packets are generated automatically by DNS and many applications.

Apply filter

```
udp
```

Observe:

- Source Port
- Destination Port
- Packet Length

Part F – Capture ARP Packets

Open Command Prompt.

Clear ARP Cache.

```
arp -d *
```

Now ping your gateway.

Example

```
ping 192.168.1.1
```

Apply filter

```
arp
```

Observe

ARP Request

```
Who has 192.168.1.1?
```

ARP Reply

```
192.168.1.1 is at xx:xx:xx:xx
```

Part G – Capture DHCP Packets

DHCP packets appear when the system requests an IP address.

Method:

Disable Wi-Fi.

Reconnect Wi-Fi.

Apply filter

```
dhcp
```

Observe

DHCP Sequence

```
Discover

Offer

Request

Acknowledgement
```

Part H – Capture IP Packets

Apply filter

```
ip
```

Observe

IPv4 Header

- Version
- Header Length
- TTL
- Source Address
- Destination Address
- Identification

Step 6: Stop Packet Capture

Click

```
Red Square (Stop Button)
```

Capture stops.

Step 7: Save Capture File

Click

```
File → Save As
```

Enter filename

```
NetworkTraffic.pcapng
```

Click

```
Save
```

Step 8: Open Saved File

Click

```
File → Open
```

Select

```
NetworkTraffic.pcapng
```

Packets appear again.

Step 9: Analyze Packet Details

Select any packet.

The screen is divided into three panes.

Pane 1

Packet List

Contains

- Time
- Source
- Destination
- Protocol

Pane 2

Packet Details

Shows

```
Ethernet Header

IP Header

TCP Header

Application Layer
```

Pane 3

Packet Bytes

Displays packet data in

- Hexadecimal
- ASCII

RESULT:

EXPERIMENT-15

To Analyze the different types of servers using Webalizer tool

Aim

To install, configure, and use the Webalizer tool to analyze web server log files and generate statistical reports for different types of web servers.

Software Requirements

- Operating System: Ubuntu Linux 20.04/22.04 (or any Linux distribution)
- Apache Web Server / Nginx Web Server
- Webalizer Tool
- Terminal
- Web Browser (Firefox/Chrome)

Theory

Webalizer is an open-source web log analysis tool used to generate graphical and statistical reports from web server log files. It processes access logs generated by web servers and provides detailed information about website usage.

The generated reports include:

- Number of visitors
- Number of page views
- Hits
- Files downloaded
- HTTP Status Codes
- Referring Sites
- Search Keywords
- Browser Statistics
- Operating System Statistics
- Hourly/Daily/Monthly Access Reports

Webalizer supports log files from:

- Apache HTTP Server
- Nginx Web Server
- IIS (Microsoft Internet Information Services)
- FTP Servers
- Proxy Servers (with supported log formats)

Step 1: Update the System

Open the Terminal and update the package repository.

Bash

```
sudo apt update
```

Upgrade installed packages.

Bash

```
sudo apt upgrade -y
```

Step 2: Install Apache Web Server

Install the Apache web server.

Bash

```
sudo apt install apache2 -y
```

Verify the installation.

Bash

```
apache2 -v
```

Check whether the Apache service is running.

Bash

```
sudo systemctl status apache2
```

If the service is not running, start it.

Bash

```
sudo systemctl start apache2
```

Enable Apache to start automatically after reboot.

Bash

```
sudo systemctl enable apache2
```



Step 4: Generate Web Traffic

To create log entries:

1. Refresh the Apache homepage several times.
2. Open different pages on the website.
3. Access the website from another computer (optional).

Each request is stored in the Apache access log

Step 5: Locate the Apache Log File

View the log directory.

Bash

```
cd /var/log/apache2
```

List the available log files.

Bash

```
ls
```

Typical output:

```
access.log
error.log
other_vhosts_access.log
```

Display the latest entries in the access log.

Bash

```
tail access.log
```

Step 6: Install Webalizer

Install Webalizer using the package manager.

Bash

```
sudo apt install webalizer -y
```

Verify the installation.

Bash

```
webalizer -V
```

The installed version number should be displayed.

Step 7: Configure Webalizer

Open the configuration file.

Bash

```
sudo nano /etc/webalizer/webalizer.conf
```

Modify or verify the following entries:

```
LogFile /var/log/apache2/access.log
OutputDir /var/www/html/webalizer
HostName localhost
Incremental yes
```

Save the file.

Exit the editor.

Step 8: Create the Output Directory

Create a directory for storing HTML reports.

Bash

```
sudo mkdir -p /var/www/html/webalizer
```

Provide appropriate permissions.

Bash

```
sudo chmod 755 /var/www/html/webalizer
```

Step 9: Generate the Report

Execute Webalizer.

Bash

```
sudo webalizer
```

If no configuration file is used, specify the log file and output directory explicitly.

Bash

```
sudo webalizer -o /var/www/html/webalizer /var/log/apache2/access.log
```

The tool reads the access log and generates HTML reports.

Step 10: View the Generated Report

Open a web browser.

Enter:

```
http://localhost/webalizer
```

or

```
http://<Server-IP>/webalizer
```

The Webalizer dashboard will be displayed.

Step 11: Analyze the Reports

The report contains the following information:

Monthly Statistics

- Total Hits

- Files Downloaded
- Pages Visited
- Visits
- Data Transferred

Daily Statistics

- Daily Visitors
- Daily Hits
- Daily Bandwidth Usage

Hourly Statistics

Shows website traffic for each hour of the day.

Browser Statistics

Displays browsers used by visitors, such as:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

Operating System Statistics

Displays operating systems used by visitors:

- Windows
- Linux
- macOS
- Android
- iOS

Referrer Statistics

Lists websites that directed users to your website.

Examples:

- Google Search
- Bing
- Yahoo

Search Keyword Statistics

Displays search terms used to access the website.

HTTP Status Code Analysis

Shows response codes such as:

Status Code	Meaning
200	OK
301	Moved Permanently
302	Found (Temporary Redirect)
403	Forbidden
404	Page Not Found
500	Internal Server Error

Top Requested Pages

Displays the most frequently accessed web pages

Example:

```
index.html
about.html
contact.html
images/logo.png
```